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Integrated decision support for promoting crop rotation based sustainable agricultural management using geoinformatics and stochastic optimization

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ABSTRACT

Sustainable agricultural management is essential for ensuring food security and economic development. Efficient agricultural land use based on crop rotation practices can deliver greater soil fertility and higher economic potential. We proposed a decision support tool (DST) for preserving land fertility, maximizing agricultural profit, minimizing agricultural pollution, and water usage. The proposed DST links geoinformatics, stochastic pairwise comparison (SPC), and constraint optimization to suggest the suitable crops for growing. To demonstrate the proposed DST, suitability of seven major crops in Muzaffarnagar district in Uttar Pradesh (India), where the footprint of sugarcane cultivable region is nearly 90% is analyzed and the findings are presented. The crops cultivated in the study region and the criteria suitable for their cultivation are identified using the hybrid system approach. The DST primarily encompasses qualitative and quantitative analysis coupled with geospatial analysis. Qualitative analysis guides the decision-maker in finalizing the crucial criteria to be assessed for cultivation, while quantitative analysis uses beta distribution for pairwise comparison to understand the significance of finalized criteria. We collected the data concerning parameters related to the finalized criteria by considering 2700 soil samples. Data required at the ungauged locations are estimated using the kriging interpolation technique. The findings of this study suggest that sugarcane can be allocated up to 20% of the land area. In addition to the principal crops (i.e., sugarcane, wheat, and rice), potato, mustard, maize, and sorghum also have good cultivation potential in Muzaffarnagar and can be grown on up to 20%, 22%, 18%, 21% of the land area respectively while just 1.5%, 1.8%, 0.1%, and 0% of land area, is used for their cultivation. With the prime focus on knowledge transfer from scientific studies to farmers, we used an open-source geospatial repository to develop an interactive dashboard that can fetch farmers' locations and present each crop's suitability based on optimized crop rotation practices.

1. Introduction

Land degradation is one of the severe environmental threats worldwide as it causes food insecurity and imbalances in the agroecosystems (Chasek et al., 2019; Prăvălie et al., 2021). Scientific surveys reveal that unsustainable agricultural practices adopted by the farmers are the prime cause of the degradation of land fertility. The United States' response to agricultural land use management includes pollution restrictions like the United States Environmental Protection Agency's Total Maximum Daily Loads (USEPA, 2010) and adoption of Best Management Practices (BMP) (Jiang et al., 2020). European Union (EU) is also examining the prospects of the Functional Land Management (FLM) concept, which aims at appropriate land-use selection based on the potential of respective soils (Schulte et al., 2015; Jiang et al., 2021).

BMPs are unanimously recognized (Wang et al., 2019; Opoku-Kwanowaa et al., 2020) for sustaining agricultural land's productivity

Abbreviations: DST, Decision Support Tool; SPC, Stochastic Pairwise Comparison; BMP, Best Management Pratices; EU, Europian Union; FLM, Functional Land Management; AHP, Analytical Hierarchy Process; GIS, Geographic Information System; DM, Decision Maker; GDP, Gross Domestic Product; NDVI, Normalized Difference Vegetation Index; ICAR, Indian Council of Agricultural Research; CPCB, Central Pollution Control Board; S1, Highly Suitable; S2, Moderately Suitable; S3, Marginally Suitable; N, Not Suitable; EC, Electrical Conductivity; OC, Organic Carbon; CO, Constraint Optimization; MIR, Maximization of Income Returns; MIWR, Minimization of Irrigation Water Requirements.

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and improving surface and groundwater quality. Crop rotation is one of the BMPs that enhances soil's organic matter, controls pests, weeds, and other diseases, prevent soil erosion, decreases dependency on chemical fertilizers, and increases crop yields (Tariq et al., 2019; Bowles et al., 2020). Past research works analyzed the cropping plan, where crop rotation decisions have been primarily dealt as a static concept, and identified the need to model these decisions as dynamic problems (Dury et al., 2012). Ridier et al. (2016) built a dynamic stochastic model to aid crop rotation planning, keeping maximization of income over the planning period as an objective with simultaneous consideration of crop yield and market risks. Li et al. (2015) proposed a heuristic algorithm for optimal crop rotation scheduling to maximize profits for smallholder farmers and minimize the profit difference between the farmers. Dos et al. (2012) suggested and analyzed a binary optimization model disintegrated using Danzig-Wolfe decomposition and a heuristic based on column generation to identify a crop rotation schedule while maximizing the plot occupation subjected to adjacency and succession constraints. Further, a crop rotation model, CropRota, is presented by (Schönhart et al., 2011), which integrates observed land-use data and agronomic criteria for maximizing the agronomic value overall rotations by assigning discrete crops to a single or multiple crop rotation. Dupuis et al. (2022) presented a six-step methodology using Markov chains to predict N most likely crops for year $n + 1$. Similarly, many mathematical or computational models have been proposed for crop rotation planning problems (Santos et al., 2015; Capitanescu et al., 2017). Although these models apply different agronomic, production, economic, market constraints, and rules, they lack consideration of individual land potentials, subjective allocation of cropping patterns to the fields, and a framework for direct field implementation of cropping plan suggestions.

Many approaches are available for evaluating land suitability. For example, the parametric approach, is a traditional method that compares crop requirements against land potentials and produces a suitability score respective to each land characteristic (Vasu et al., 2018). FAO's (1981) land evaluation framework is also an extensively used approach to determine crop-land suitability based on soil, climate, and terrain data. Bagherzadeh et al. (2016) modeled the land suitability using parametric neural networks and fuzzy logic approaches for soybean production in a semi-arid region and identified soil infertility as the manor limiting factor. Vasu et al. (2018) compared parametric, multicriteria, and Storie index land suitability evaluation approaches for assessing land suitability for pigeon pea, maize, cotton, groundnut, and rice in India. Similarly, land suitability evaluation for growing soyabean, wheat, and maize was carried out by Dedeoğlu and Dengiz (2019) and Ramamurthy et al. (2020), respectively, for wheat using a hybrid analytic hierarchy process (AHP) and geographic information system (GIS) model. Most of these works focused on the land suitability evaluation without providing an on-ground implementable decision support tool (DST) to the farmers to shift from monocropping to crop rotation practices while enhancing their income as well as soil characteristics. The monoculture cropping system hampers the crop yield and is also detrimental to soil fertility. In addition, the monoculture cropping system may cause pest growth and a decline biodiversity of soil organisms. It prevents benefits that would otherwise come from practicing crop rotation (Sahajpal et al., 2014).

For promoting crop rotation, the criteria for selecting suitable land for a specific crop belong to different subdomains concerning climate bio-physicochemical characteristics of the land. In earlier studies, the AHP and fuzzy-AHP have been extensively used for pairwise comparisons of criteria (Cobuloglu and Büyüktahtakın, 2015; Srinivas and Singh, 2018; Dedeoğlu and Dengiz, 2019; Srinivas et al., 2020). However, these approaches have certain limitations, such as rank-reversal and allowance of only some selected predefined ranges while converting linguistic variables to fuzzy sets and also an inability to offer flexibility to the Decision Maker (DM) to choose their own scale for rating the criteria. In this study, we have used a stochastic pairwise comparison approach, which utilizes beta distribution for the pairwise comparison of different criteria. Unlike the limitation of fixed ranges in conventional approaches, the DM in this approach has the flexibility to define their range limits (Cobuloglu and Büyüktahtakın, 2015).

The study proposes an integrated approach linking geoinformatics, SPC, and constraint optimization for sustainable crop rotation planning to address the intimidating issues of declining water quality and quantity, land fertility, and enhancing farmers income return based on crop rotation practices. The DST overcomes the major drawback in past crop planning models by combining the US and EU response towards agricultural land use management, i.e., extending the FLM concept in terms of delineating the suitable crop-rotation pattern (a BMP type) for the given land based on the soil's physical and chemical characteristics and climatic conditions. The availability of study results through a web-based dashboard encourages field scale implementation of cropping patterns by the farmers and the participation of other stakeholders. Seven major crops from the Muzaffarnagar district in Uttar Pradesh (India) are selected to demonstrate the proposed DST. The main objectives include 1) identification of major crops and corresponding criteria for their cultivation and evaluating their weights using hybrid system approach, 2) generating thematic layer maps followed by generation of land suitability maps specific to each of the seven major crops using food and agriculture organization (FAO) classification system, and 3) identifying optimum crop rotation pattern required for sustaining soil fertility, maintaining water quality and quantity, and enhancing farmers income. Moreover, the proposed approach's future development and universal applicability are also discussed.

2. Materials and method

The Muzaffarnagar district contributes immensely to Uttar Pradesh's (India) gross domestic product (GDP) and is known for its sugar and jaggery production. The district covers an area of about 2960 km^2 and is surrounded by the Rivers Ganges and Yamuna. It is a land of high fertility, with wheat, rice, and sugarcane being its principal crops. The geographic location of the district covers north latitudes from 29° 10′ 49.33″ to 29° 42′ 33.33″ and east longitudes from 77° 23′ 10.06′' to 78° 08′ 13.18′' (Fig. 1).

The massive sugarcane cultivation in the region has resulted in overextraction and contamination of groundwater despite reasonable average annual rainfall (870 mm). Due to the farmers' general tendency to grow sugarcane year after year to obtain maximum income, its overcultivation has degraded the land fertility. The district has about 2200 sq. km. of net sown area, primarily irrigated using tube wells and canals. According to the reports (PTI, 2021a, 2021b), the farmers grow sugarcane on 90% of the land. This high production is becoming difficult for the mills to process, causing the excess sugarcane crop laid on tractor-trollies dry in the sunlight. Such disproportion in production and demand also causes stagnancy in sugarcane prices. Thus, this study aims to provide farmers with cost-effective potential solutions based on implementing crop rotation practices for preserving land fertility, maximizing agricultural profit, minimizing agricultural pollution and water usage. The proposed framework involves the integrated application of SPC, GIS, and the collection of in-situ and web-based geospatial datasets under a hybrid system approach. This is exercised by applying a suite of subsequent analytical steps highlighted in the flow scheme in Fig. 2.

2.1. Database curation for the study

Table 1 summarizes the datasets, time period, and sources used in developing the crop-rotation planning framework. The references for data sources are listed in Supplementary material (A3 section) for brevity. Soil's physical and chemical properties have a great influence on crop yields. They directly influence the agrarian activities like erosion, tillage, irrigation, and drainage (Rogers et al., 2015). In general, 17 elements are regarded essential for plant growth and based on their high or low quantitative requirement by the crops, these elements are

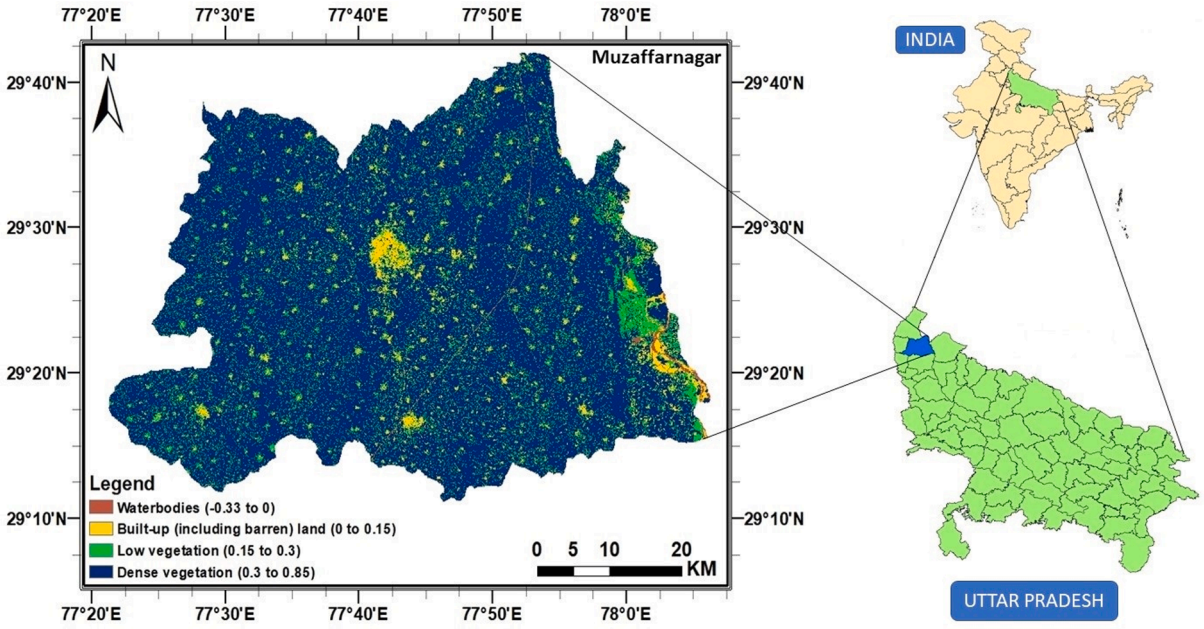


Fig. 1. Location and land use classification of the study area using NDVI for the year 2020.

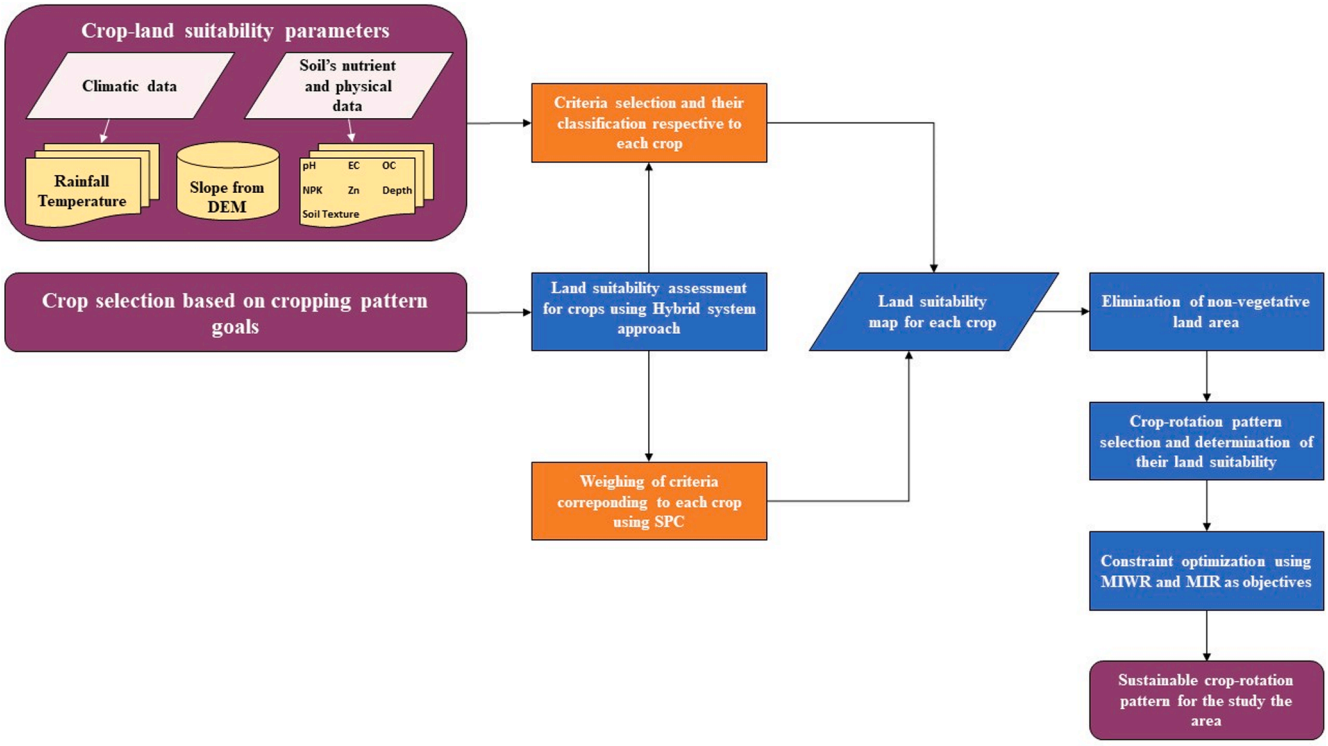


Fig. 2. Schematic representation of the adopted methodology for sustainable crop-rotation pattern estimation.

classified as macronutrients and micronutrients respectively (Parikh et al., 2012). Zolekar and Bhagat (2015) and Dedeoğlu and Dengiz (2019) are two related studies which have used soil properties for determining land suitability. The detailed information on how these properties influence crop's metabolism events can also be found in these related articles. Both categories of soil nutrients viz., macronutrients and micronutrients, are essential for plants' metabolism events. However, micronutrients are not considered in the present study except for rice, as zinc deficiency has been reported as a widespread issue by many studies particularly for the rice grown in flooded conditions (Wissuwa et al., 2006; Impa et al., 2013). The exclusion of the micronutrients owes to the various limitations related to their field-scale applicability, such as lack of subsidies and high goods and services tax on micronutrients as compared to macronutrients, lack of technical knowledge related to micronutrients at field level, limited soil testing facilities (Shukla et al., 2020), and dearth of knowledge or the active research for understanding micronutrients dynamics and transformations under different agricultural productions systems, understanding various biological and physiochemical properties of soil which affect micronutrients' uptake, utilization, and availability (Wang et al., 2016; Nadeem and Farooq, 2019).

Table 1
Summary of the dataset used for development of crop-rotation planning framework.

Data used	Source	Time period
Sentinel-2A satellite images	USGS, 2021	2020
Soil's macro-nutrient and physical parameters	SHC, 2021	2017–18 to 2018–19
Bulk density & Soil depth	ISRIC, 2021	2021
Precipitation & Temperature	NASA, 2021	Jan. 1990 to Dec. 2019
Soil Texture classification	NBSS, 2021; CGWB, 2018	2018
DEM	ISRO, 2021	2021

The data concerning the physical and chemical properties of soil and climate for the Muzaffarnagar district are obtained from various in-situ-based, web-based geospatial datasets. The in-situ collection of soil's macro-nutrient and physical parameters, e.g., Nitrogen (N), Phosphorus (P), Potassium (K), pH, Electrical conductivity (EC), etc., from 15 to 20 cm depth was used for preparing thematic layers for land suitability estimation. The bulk density layer map for 15–30 cm depth and the soil depth information was used for the standardization of soil macronutrient data. The 30-year monthly annual average precipitation data from January 1990 to December 2019 averaged over the months of two cropping seasons, i.e., Rabi (October to April) and Kharif (May to October) were also used. The average air temperature 2 m above the earth surface has also been attained, where the data obtained for random discrete points were interpolated using ArcGIS (ver. 10.9). Further, the soil texture classification layer maps for the study region were created using ArcGIS from the soil texture classification maps published by the National Bureau of Soil Survey and Land Use Planning (NBSS, 2021) and a groundwater management report (CGWB, 2018). Other important datasets used were CartoDEM: Version-3 tiles encompassing the study area, Sentinel-2A satellite images with the cloud over less than 10% for preparing Normalized Difference Vegetation Index (NDVI) maps for identifying the cultivable land area of the selected region.

2.2. Criteria selection, classification, and thematic layers preparation

We have used a hybrid system approach for evaluating the land suitability for selected crops. Two model types, i.e., quantitative and qualitative, are assembled for land suitability evaluation. The opinions obtained from stakeholders (farmers, experts from the Indian Council of Agricultural Research (ICAR), Central Pollution Control Board (CPCB), and Department of Irrigation) constitute the qualitative model. Qualitative model is used for selecting crucial parameters concerning soil's physical, chemical and climatic properties for each crop (Table 2). Four classes are created for each criterion based on land evaluation classification system as exercised by FAO (1996), United Nations. These classes are S1 (highly suitable), S2 (moderately suitable), S3 (marginally suitable), and N (not suitable) for each of the selected crops.

Table 2
Land suitability parameters for each crop.

Crop type	Soil's physical properties	Soil's chemical properties	Climatic properties
Paddy	Texture, Slope	pH, EC, N, P, K, OC, Zn	Temperature, Rainfall
Wheat	Texture, Slope, Depth	pH, EC, N, P, OC	Temperature, Rainfall
Maize	Texture, Slope	pH, EC, P, K, OC	Temperature, Rainfall
Mustard	Texture, Slope	pH, EC, N, P, K, OC	Temperature
Potato	Texture	pH, EC, OC	Temperature, Rainfall
Sugarcane	Texture, Slope, Depth	pH	Temperature, Rainfall
Sorghum	Slope, Depth	pH, EC, N, P, OC	Temperature, Rainfall

Quantitative modeling uses stochastic pairwise comparison (SPC) for each crop, which determines the relative weights of each criterion in terms of their importance in the cultivation of the crop under discussion. Based on the classification system and curated database discussed earlier, the thematic map preparation for each criterion specific to each crop type was carried out using the ordinary kriging interpolation technique, where the spherical semivariogram model was used for modelling the empirical semivariogram (Yakushev et al., 2020). Consequently, the thematic layers for criteria were prepared by classifying and standardizing prepared raster maps into four suitability criteria specific to each crop using the raster calculator tool in the GIS environment. The importance score for each class (i.e., S1, S2, S3, and N) is established through conventional AHP technique (Saaty, 1980; Ramamurthy et al., 2020). The resultant eigenvector weights with acceptable consistency ratio [i.e., 0.06 $<$ 0.1], were 0.5666, 0.2674, 0.1667, and 0.0399 corresponding to S1, S2, S3, and N class respectively.

2.3. Weighting of criteria using stochastic approach

The SPC scheme evaluates the relative weights of parameters suitable for each crop. To convert DM's valuation into a crisp value, a closed-form approximation of the median for the β-distribution is used. The novelty of this approach is that instead of accepting only a crisp number from the experts when comparing two criteria, the approach provides flexibility to the experts to present their opinion either in the form of a 'range of values' or 'most probable value with a lower and upper bound' or a 'crisp comparison'. Further, SPC deals with the uncertainty or imprecision associated with DM's choice by modeling uncertain comparisons to stochastic pairwise distributions. The approach terminates with successful consistency attainment in a pairwise comparison matrix. A detailed description of this approach is as follows:

- Construction of pairwise comparison matrix specific to each crop: DM performs comparison based on crisp responses, most probable values with lower and upper bounds, or by using a range of values. For illustration, according to qualitative model, for potato cultivation, if soil pH is moderately more important than organic content in the soil, then the score will be '$P_{Po} = 3$'; if judgment says that temperature is either equally or moderately more important than slope for rice cultivation, then the input will be $R_{ts} = (1, 3)$; or in case of wheat, if the texture is at least in between equally and moderately important, most likely it is in between moderately and strongly important, or at most strongly more important than pH, then we will obtain $W_{tp} = (2, 4, 5)$.
- Converting stochastic pairwise comparison values are into crisp values: If 'l' and 'm' are the two criteria related to crop A, then the crisp value for the corresponding pairwise comparison is equal to A_{lm}. And let $G_{lm}(A_{lm}|\Phi_{lm})$ be the probability density function with parameters Φ_{lm} for converting stochastic comparison values to crisp. Following the previous step, for example, $W_{tp} = (2, 4, 5)$ can be modelled using a triangular distribution with lower limit (ll), most likely (ml), and upper limit (ul) as $W_{tp}\ \ G_{tp}(\mathrm{ll}, \mathrm{ml}, \mathrm{ul}) = t_{tp}(2, 4, 5)$. Similarly, $R_{ts} = (1, 3)$ can be modelled using a uniform distribution function, $R_{ts}\ \ G_{ts}(\mathrm{ll}, \mathrm{ul}) = u_{ts}(1, 3)$ and the direct crisp comparisons are simply written as $P_{po}\ \ G_{po} = 3$. For standardization of different distributions, a pairwise β-distribution function is used, $\beta(\Theta \mid \upsilon, \omega, \mathrm{ll}, \mathrm{ul})$ where Θ is beta-distributed pairwise comparison value, (υ, ω) are the shape factors, and (ll, ul) are the location parameters, with $\mathrm{ll} \leq \Theta \leq \mathrm{ul}$ and $\upsilon, \omega \geq 1$. The methods of moments (MoM) are used to model all A_{lm} to β-distributed functions with shape factors $(\upsilon_{lm}, \omega_{lm})$ and location parameters $(\mathrm{ll}_{lm}, \mathrm{ul}_{lm})$ (Jalao et al., 2014). Using the first and the second moment (Eq. (1) and Eq. (2), respectively), sample

mean and variance, as represented in Eq. (3) and Eq. (4), are calculated.

$$E\left[\widetilde{A}\right] = ll + \frac{\upsilon}{\upsilon+\omega}(ul-ll) \tag{1}$$

$$Var\left[\widetilde{A}\right] = \frac{\upsilon\omega}{(\upsilon+\omega)^2(\upsilon+\omega+1)}(ul-ll) \tag{2}$$

where,

$$E\left[\widetilde{A}\right] = \frac{ll+ml+ul}{3} \tag{3}$$

$$Var\left[\widetilde{A}\right] = \frac{ll^2+ml^2+ul^2-llml-llul-mlul}{18} \tag{4}$$

Equating Eqs. (1) & (2) to Eqs. (3) & (4) respectively, υ and ω (Eqs. (5) and (6)) are obtained as follows:

$$\upsilon = \left(\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\left(\frac{\left(\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\left(1-\left(\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\right)}{Var\left[\widetilde{A}\right]/(ul-ll)^2}-1\right) \tag{5}$$

$$\omega = \left(1-\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\left(\frac{\left(\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\left(1-\left(\frac{E\left[\widetilde{A}\right]-ll}{ul-ll}\right)\right)}{Var\left[\widetilde{A}\right]/(ul-ll)^2}-1\right) \tag{6}$$

The outputs for conversion of SPC to β-distributed pairwise comparison after applying MoM are as mentioned in the Eqs. (7) to (9) below:

$$\Theta_{lm} \quad \beta(\upsilon_{lm},\ \omega_{lm}, ll_{lm}, ul_{lm}) \text{ if } A_{lm} \quad t(ll_{lm}, ml_{lm}, ul_{lm}) \tag{7}$$

$$\Theta_{lm} \quad \beta(\upsilon_{lm}=1,\ \omega_{lm}=1, ll_{lm}, ul_{lm}) \text{ if } A_{lm} \quad u(ll_{lm}, ml_{lm}, ul_{lm}) \tag{8}$$

$$\Theta_{lm} \quad A_{lm} \text{ if } A_{lm} \text{ is crisp} \tag{9}$$

More details on converting SPC to β-distributed pairwise comparison are given in Jalao et al. (2014).

- The closed-form approximation as developed by (Kerman, 2011), is used to convert β-distributed pairwise comparison to crisp values. The crisp value is obtained using the median of β-distribution (Eq. (10)),

$$M(\upsilon_{lm},\ \omega_{lm}) \approx \frac{\upsilon_{lm}-1/3}{\upsilon_{lm}+\omega_{lm}-2/3} \tag{10}$$

subjected to constraints as depicted in Eq. (11),

$$\frac{\upsilon_{lm}-1}{\upsilon_{lm}+\omega_{lm}-2} \le M(\upsilon_{lm},\ \omega_{lm}) \le \frac{\upsilon_{lm}}{\upsilon_{lm}+\omega_{lm}} \tag{11}$$

Eq. (11) is applicable for the cases when $\upsilon_{lm} \ge \omega_{lm}$, elsewise the inequalities would get reversed. Lastly, the crisp value for Θ_{lm} is attained using Eq. (12) as defined below:

$$\Theta_{lm} = ll_{lm} + (M(\upsilon_{lm},\ \omega_{lm})^*(ul_{lm}-ll_{lm})) \tag{12}$$

After constructing a pairwise comparison matrix for all criteria, the weight or value of each criterion for supporting the growth of a particular crop is calculated using a priority eigenvector followed by consistency evaluation.

2.4. Predicting sustainable crop rotation pattern

The criteria weights and the respective thematic map layers with four classified intervals for a particular crop were overlaid in ArcGIS to obtain a suitable land pattern for each crop. The resulted crop-land suitability maps were then overlaid with the NDVI map for the year 2020 to produce crop-land suitability maps suggesting the suitable land only for the region which supports vegetation, i.e., the areas with built-up (including barren) land and waterbodies have been eliminated. The resulted crop-suitability land areas were further used for estimating optimal crop rotation pattern using constraint optimization (CO) based on the two objectives, i.e., maximization of income returns (MIR) and minimizing irrigation water requirements (MIWR). Table 3 gives detailed information on irrigation water requirements and net income return for each cropping pattern.

These cropping patterns are the different possible combinations of the seven selected crops and are obtained through past agricultural experiences in the study area, experts' consent, and detailed literature surveys (Chandrasekaran et al., 2010; Parihar et al., 2011). In addition to CO crop-rotation pattern, flexible cropping patterns corresponding to each objective were derived for addressing subjective needs of the farmers.

3. Results

3.1. Preparation of classified thematic maps for crops and relative weights using SPC

The criteria selection for the seven selected crops is accomplished by applying the 'Hybrid system approach'. The list of criteria with suitability classification (i.e., S1, S2, S3, and N) for rice and wheat are provided in Table 4 and Table A1 (Supplementary material), respectively. Equivalent data corresponding to other crops is also created. Using this data and the land suitability parameters (Fig. A2), the classified thematic maps for each criterion corresponding to rice crop (Fig. 3) and other crops were prepared.

Now, the quantitative modeling results for the rice crop, as an

Table 3
Crop-rotation patterns and corresponding information on irrigation water requirements and net income return.

Sr. No.	Crop-rotation patterns	Irrigation water requirements (mm/annum)	Net income return (₹/hectare/annum)
1	Rice-Wheat-Fallow	750	28189.00
2	Maize-Wheat-Fallow	800	22167.33
3	Maize-Potato-Maize-Wheat-Fallow	980	48470.40
4	Maize-Mustard-Fallow	700	21464.80
5	Sorghum-Mustard-Fallow	633.33	66177.47
6	Sorghum-Wheat-Fallow	733.33	23878.67
7	Rice-Mustard-Fallow	650	28486.47
8	Rice-Wheat-Sugarcane-Fallow	1562.5	68108.25
9	Maize-Wheat-Sugarcane-Fallow	1600	62842.00
10	Rice-Potato-Sugarcane-Wheat-Fallow	1490	85485.40
11	Rice-Mustard-Sugarcane-Wheat-Fallow	1410	63194.28
12	Rice-Wheat-Sugarcane-Wheat-Fallow	1470	63615.80
13	Rice-Potato-Maize-Potato-Fallow	970	74553.00
14	Maize-Potato-Sugarcane-Fallow	1625	90179.00

Sources: (FAO, 1986; MAFW, 2018; DAC, 2019).

Table 4
Suitability range for criteria corresponding to rice.

Land suitability parameters ↓	S1	S2	S3	N
Suitability class score →	0.5667	0.2674	0.1667	0.0399
Temperature (°C)	25 to 30	30 to 35	10 to 20	<10 and > 35
pH	5.5 to 7.5	7.5 to 8 5 to 5.5	8 to 8.5 4.5 to 5	>8.5 <4.5
Precipitation (mm)	>1500	1000 to 1500	700 to 1000	<700
Electrical Conductivity (dS/m)	<3	3 to 5	5 to 7	>7
Soil Texture	Loam, Sandy clay, and Clay	Sandy Clay, Silty loam, Silty clay	Silt, Sandy Loam	Gravel, Sand
Slope (%)	<1	1 to 2	2 to 6	>6
Nitrogen (ppm)	>80	80 to 40	40 to 20	<20
Phosphorus (ppm)	>15	15 to 10	10 to 5	<5
Potassium (ppm)	>400	400 to 200	200 to 100	<100
Organic Carbon (%)	>2	2 to 1	2 to 0.5	<0.5
Zinc (ppm)	>1	1 to 0.5	0.5 to 0.25	<0.25

example, out of the seven crops are discussed, and the relative weights of the soil properties, topography, and weather criteria are determined. A total of 11 parameters are considered, and stakeholders are consulted, and their judgments (comparison matrix) are summarized in Table A2 (Supplementary material). This comparison matrix (Table A2) is transformed into a β-distributed pairwise comparison using equations (7–9), and the resulted matrix is given in Table 5. The matrix with crisp comparison (Table A3) is obtained using equations (10–12). Finally, the relative weights of the parameters for rice cultivation are presented in Table 6. The consistency ratio value of 0.099 indicates that the matrices are consistent (Ramamurthy et al., 2020). Similarly, the parameters' weights of other crops are also determined.

Table 6 indicates that temperature (17.45%), rainfall (16.94%), soil texture (13.16%), and slope (13.51%) are the major parameters influencing rice cultivation, followed by electrical conductivity (9.32%), pH (8.42%), and Nitrogen (7.56%) (Ran et al., 2018; Ujoh et al., 2019).

3.2. Procurement of land suitability patterns for Muzaffarnagar

The classified thematic layers prepared earlier corresponding to each criterion for each crop were overlaid with the help of the weights obtained using ArcGIS to derive land suitability maps for all the crops (Fig. 4).

The land use classes corresponding to built-up (including barren) areas and water bodies have been removed from the map and the resulted vegetative land area is obtained as shown in Fig. A1. Thus, we eliminated the non-vegetative land area from the land suitability maps of all seven crops. The eliminated area accounts for around 2.5% (6,950 ha) of the total area. The resultant land suitability crop maps indicate that only 3.5% (9,015 ha) of land constitutes high suitability for rice cultivation in Muzaffarnagar while about 26% (67,400 ha) of land lying mostly in the upper region of the district (i.e., covering portions of Charthawal, Muzaffarnagar, Morna, and Purkazi blocks) supports moderate and 68% (177,210 ha) of the remaining land supports marginal cultivation of rice. The sugarcane crop partakes maximum percentage under high suitability region compared to other crop i.e., 60% (150,035 ha) under the high suitability region and about 40% (1,02,341 ha) under the moderate suitability region. Similarly, for the wheat crop 31% (798,23 ha) land is under high suitability region whereas 66% (1,71,835 ha) land is under moderate suitable region.

Interestingly, results indicate that potato, maize, and mustard crops also have high cultivation potential in Muzaffarnagar district, with around 20% (66,361 ha) land for mustard, and 25% (64,500 ha) for both

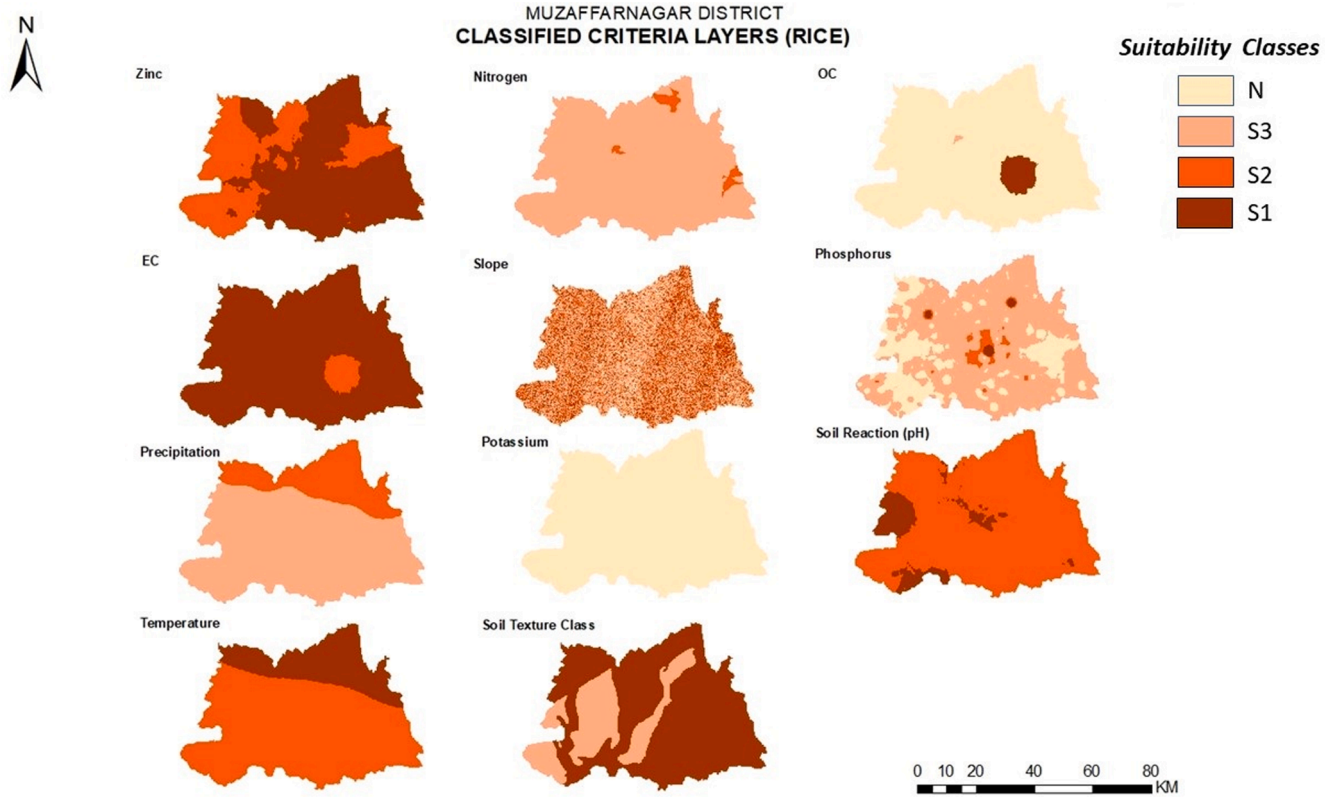


Fig. 3. Classified thematic maps for selected criteria corresponding to Rice crop.

Table 5
β-distributed pairwise comparison matrix for rice.

Criterion	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11
Temperature (C1)	1	β (1,1,1,3)	β (1,1,1,3)	β (2.5,2.5,2,4)	β (2.09,2.62,2,5)	β (1,1,1,3)	β (1,1,1,3)	β (2.62, 2.09,2,5)	β (2.5,2.5,2,4)	β (2.62, 2.09,2,5)	β (2.5,2.5,2,4)
pH(C2)		1	β (1,1, 1/ 5,1/4)	β (2.5,2.5,1,3)	β (2.5,2.5,1/6,1/ 3)	β (2.58, 2.37,1/ 7,1/3)	β (1,1,2,3)	3	3	3	3
Rainfall(C3)			1	β (2.5,2.5,2,4)	β (2.5,2.5,2,4)	3	2	β (2.09, 2.62,2,5)	β (2.62, 2.09,2,5)	β (2.5,2.5,2,4)	β (2.09, 2.62,2,5)
EC(C4)				1	β (1.66, 2.48,1/6,1/3)	β (1.66, 2.48,1/ 6,1/3)	β (2.5,2.5,2,4)	β (2.5,2.5,3,5)	β (2.09, 2.62,3,6)	β (1.8269, 2.5577,3,7)	(4)
Texture(C5)					1	β (1,1,2,3)	β (1,1,1,2)	3	3	3	β (1,1,3,4)
Slope(C6)						1	β (1,1,2,3)	4	4	4	β (1,1,4,5)
N(C7)							1	3	3	3	β (1,1,3,4)
P(C8)								1	1/2	1/2	1
K(C9)									1	1	2
OC(C10)										1	2
Zn(C11)											1

Table 6
Land suitability criteria for rice cultivation and corresponding weights.

Criterion	Weights
Temperature (C1)	0.1745
pH(C2)	0.0842
Rainfall(C3)	0.1694
EC(C4)	0.0932
Texture(C5)	0.1316
Slope(C6)	0.1351
N(C7)	0.0756
P(C8)	0.0291
K(C9)	0.0391
OC(C10)	0.0388
Zn(C11)	0.0293

Consistency index (CI) = 0.1500, RI = 1.51, Consistency ratio = 0.099

maize and potato lie within high suitability (S1) region with 35% to 45% land moderately good for cultivation of these crops. Sorghum crop also found to have a good cultivation potential in Muzaffarnagar.

3.3. Outcomes of crop-rotation planning and optimization

This section presents the results of crop-rotation planning for Rabi and Kharif seasons using the seven crops (as discussed above) based on the two objectives (MIR and MIWR). Crop rotation planning is accomplished by categorizing the crop-land suitability areas into two classes namely 'suitable land' and 'non-suitable land'. The land area under 'S1′ and 'S2′ region (Fig. 4) is classified as 'suitable land' and 'S3′ land area as 'non-suitable land'. Using stakeholder's opinions on seven selected crops, different agronomically feasible permutations of crop-rotation patterns were formed as summarized in Table 3. The 'suitable/non-suitable land' information for each crop is then overlaid and intersect respective to each crop-rotation pattern, and the land area suitable for all crop-rotation patterns is deduced. Now, a sustainable or optimized crop-rotation pattern is estimated for each land parcel in the entire Muzaffarnagar district using MIR and MIWR objective functions (Fig. 5). Additionally, the crop-rotation patterns corresponding to only MIWR (Fig. A3) and only MIR (Fig. A4) objective were also suggested to provide flexibility to the farmers and the local authorities for choosing a particular cropping pattern as per the native absolute requirement.

The annual irrigation water consumption, annual net return, and total acreage corresponding to each cropping pattern for each of the three aforementioned scenarios are summarized in Tables 7, 8, and Table A4.

Essentially, the CO scenario is a trade-off between the two non-complementary objectives, MIWR and MIR. Following the results (Table 7, 8, and Table S4), the CO scenario provides 50% more income returns on the cost of 30% more water than the MIWR scenario. On the other hand, the crop rotation patterns in CO ensures 20% less profit than the maximum profit (MIR) scenario but save nearly 40% of water consumed.

Results suggest that the annual average cultivation or incorporation of (18%) maize and (20%) potato crops in agricultural fields of Muzaffarnagar in addition to (20%) sugarcane could provide maximum net income returns to the farmers while practicing crop rotation (Fig. 6).

Whereas as exclusion of these crops support conservation of irrigation water. A trade-off of between these two opposing instances is achieved via CO scenario. On the other hand, cultivation of (22%) mustard and (21%) sorghum crops on the Muzaffarnagar land parcels on the suggested locations is highly conducive for water conservation. Also, as can be inferred from Fig. 6, a minimum of 25% of the cultivated land area should be allocated to fallow lands while practicing crop-rotation patterns.

4. Discussion

4.1. Flexible relative criteria weights determination using SPC approach

The SPC approach applied for identifying the relative weights of criteria (Table 6) excels the conventional AHP technique in offering flexibility to the decision-maker without compromising the quality of results as the results obtained are consistent with the findings of Ujoh et al. (2019) and Amini et al. (2020), where they have used the conventional AHP approach. Also, the dominance of temperature over other parameters (Table 6) is supported by the former studies (Xiong et al., 2009; Liu et al., 2014). It could be explained due to the direct effect of temperature on the paddy yield (Liu et al., 2014). Temperature is also a major factor concerning photosynthesis and formation of carbohydrates which is the primary constituent of rice (Xiong et al., 2009). The temperature difference also influences heat stress (Lobell, 2007) which in turn affects the soil respiration and nutrient absorption in rice (Xu et al., 2015). Rice being a tropical crop, requires high heat and humidity for its healthy growth. Consequently, temperature and rainfall parameters

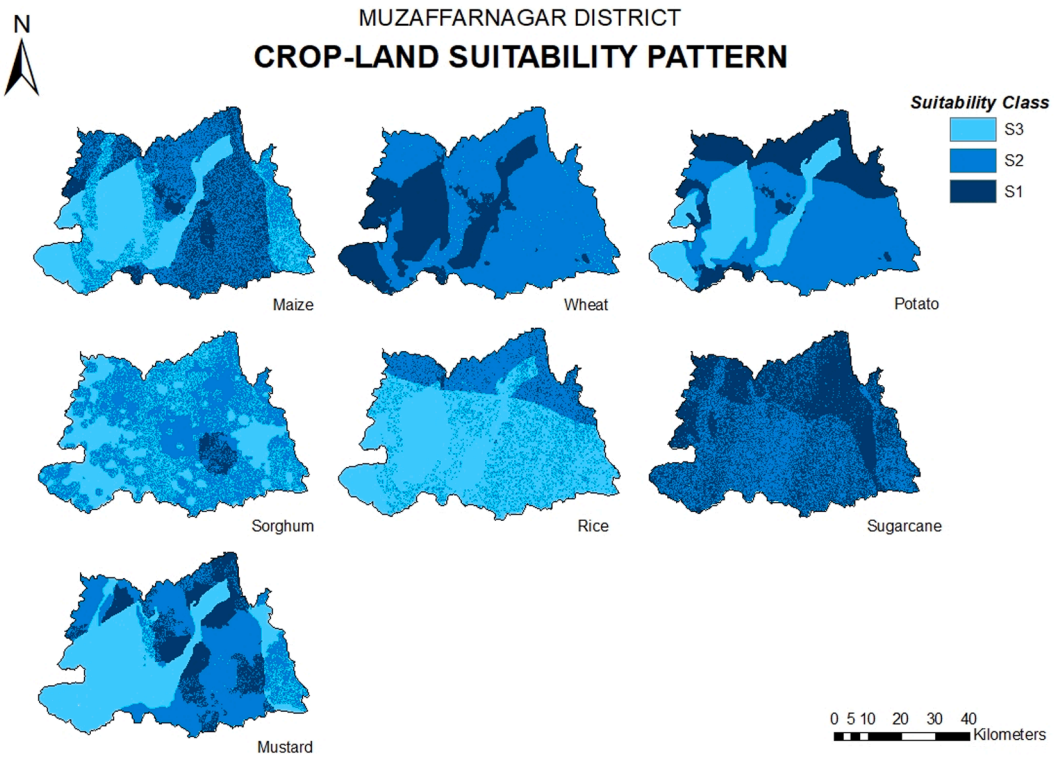


Fig. 4. Land suitability maps for crops for Muzaffarnagar district.

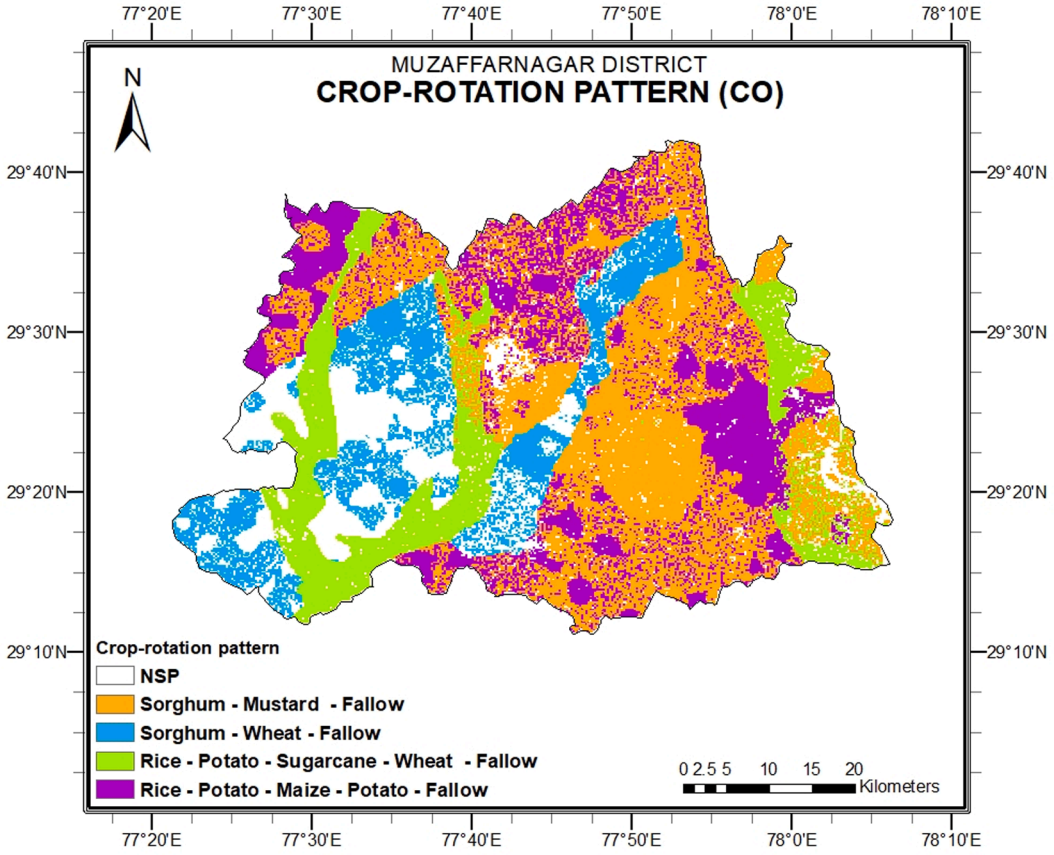


Fig. 5. Crop-rotation pattern for Muzaffarnagar district corresponding to the Constraint Optimized (CO) scenario.

Table 7
Net return, total water consumption, and acreage for the crop-rotation patterns under the constrained optimized scenario.

Constraint optimization (CO)	Acreage (ha × 1000)	Total Water Consumption (Mm^3)	Net Return (Million $)
Sorghum - Mustard - Fallow	90.48	573.06	80.46
Sorghum - Wheat - Fallow	33.26	243.89	10.67
Rice - Potato - Sugarcane - Wheat - Fallow	38.68	576.39	44.43
Rice - Potato - Maize - Potato - Fallow	53.76	521.48	53.86
NSP	44.53	0	0
Total	260.72	1914.82	189.43

Table 8
Net return, total water consumption, and acreage for the crop-rotation patterns under minimum irrigation water scenario.

MIWR	Acreage (ha × 1000)	Total Water Consumption (Mm^3)	Net Return (Million $)
Rice - Wheat - Fallow	26.95	202.16	10.57
Sorghum - Mustard - Fallow	90.48	573.06	80.46
Sorghum - Wheat - Fallow	44.99	329.90	14.43
Rice - Mustard - Fallow	53.76	349.45	20.58
NSP	44.53	0	0
Total	260.72	1454.57	126.05

consume maximum percentage for supporting rice growth (Ran et al., 2018). After meeting the fundamental and essential climatic requirements, soil and terrain characteristics influence the rice culture maximum, which could be explained based on the fact that both soil texture and slope are directly related to soil's water holding capacity. No doubt, the nitrogen is a leading nutrient among others as it improves panicle count, stomatal conductance, and facilitates photosynthesis and transpiration, however, it is overruled by the EC and pH parameters. It could be true because the non-conducive pH and EC environments could impede the nutrient absorption of plants (Huang et al., 2017).

4.2. Scientific agricultural land-use allocation using proposed DST

Muzaffarnagar is popularly known as the sugar bowl of India, the crop land-use pattern maps (Fig. 4) obtained using the proposed DST also propound the concurrent statistics with sugarcane having the highest high-land suitability region. The suitable sugarcane growth requirements encompass about neutral soil pH (6.5 to 7.5), loamy to sandy loamy soil texture, good soil depth, 20 to 30 °C temperature, and rainfall greater than 950 mm (Rasheed and Venugopal, 2009; ICAR, 2021) in the study area are the potential factors for the observed trend. Rice being a temperature sensitive and water loving crop, requires more than 1500 mm precipitation and around 25 °C to 30 °C temperature for its optimum growth. However, the highest annual average precipitation being 1050 mm and the high average temperature (>29.5 °C) in the study region could be a cause of low high-suitability paddy farming regions (ICAR, 2021). Similarly, the upper portions, rich in precipitation with lower temperature ranges, are more suitable for paddy cultivation than the remaining parts of the district. The results obtained are congruent with the general observed trend i.e., wheat and sugarcane crops are the two most suitable crops for the land in Muzaffarnagar, with rice being the third principal crop with relatively low production (KVK, 2015).

The observed high cultivation potential of potato, maize, and mustard crops is primarily due to the favourable slope (i.e., >90% region with slope between 0 and 6%), soil texture (loam and sandy loam), and rainfall conditions (>800 mm) for maize (Tashayo et al., 2020); favourable soil texture (loam and sandy loam), and phosphorus content (10 to 55 Kg/hectare) for mustard (Mandal et al., 2020); and texture (loam and sandy loam), and rainfall (800 mm or more) are the most favourable conditions for potato cultivation (Mandal et al., 2020; Sangeeta, 2021). However, as summarized by agricultural reports (KVK, 2015; MAWF, 2018), potato, mustard, maize, sorghum crops are cultivated only on 1.5%, 1.8%, 0.1%, and 0% land area.

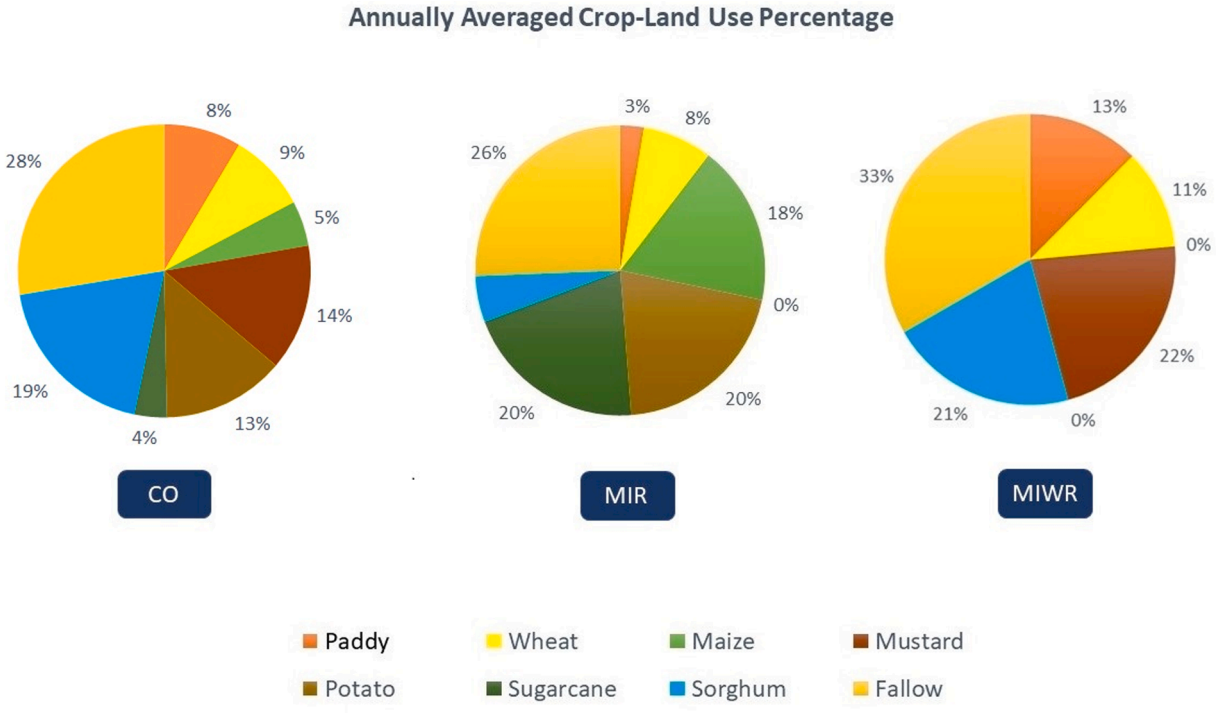


Fig. 6. Annual average percentage of land covered by crops under MIR, CO, and MIWR scenario.

Thus, some crops with good cultivation potential are not given due consideration. This could be because of the lack of land use and crop rotation planning, leading to biased cropping selection by the farmers like considering the market or economic aspects alone, which may sustain for some short period of time (ICAR, 2021). But for sustainable cropping pattern selection, a comprehensive consideration is necessary, i.e., simultaneous achievement of ecological, hydrological, and socio-economic aspects (Li et al., 2015). Sugarcane is the most suitable crop for Muzaffarnagar, thus, naturally it has a good market. The lack of proper land use planning has led to extensive cultivation of sugarcane, and consequently, complications have risen, such as overexploitation of the groundwater, land degradation, high usage of chemical fertilizers, and impaired water quality. According to Central Ground Water Board (CGWB, 2018), the groundwater in three out of nine blocks (Baghra, Budhana, Charthawal) in the Muzaffarnagar district is found to be overexploited, with one block (Shahpur) reaching a critical state. Sustainable crop rotation planning is one of the viable solutions which can account for economic, environmental, agronomical, hydrological considerations (Tariq et al., 2019).

4.3. Optimized crop-rotation patterns for environmental and economical benefits

The allocation of crops based on CO, MIWR, MIR (Fig. 5, A3, and A4) scenarios provides substantial benefits like providing a natural pest control system, soil hygiene, enhancement in crop yield, minimizing fertilizer usage, etc., by preventing overexploitation of soil because of excessive cultivation of one crop-type. Eventually, it would preserve soil fertility and prevent water degradation by limiting the use of pesticides and insecticides in the field (Bonanomi et al., 2020; CEF, 2021). In addition to the aforementioned factors, the sustainable or constraint optimized cropping pattern is an attempt to provide maximum and long-term benefit to farmers while simultaneously minimizing water consumption. The supplementation of three high profitable crops, i.e., maize, potato, and sugarcane (Islam and Islam, 2018; Wei et al., 2020), encompassing over 20% land area, to the CO scenario in comparison with MIWR could explain the prevalent high income returns in CO scenario. While the reduction in high income potato and sugarcane crops (23%) (MAFW, 2018) and addition of low water requiring sorghum and mustard crops (28%) (FAO, 1986; Assefa et al., 2010; Shekhawat et al., 2012) to CO scenario in comparison with MIR provide a rationale for the observed variation in MIR and CO scenario. According to the general agricultural statistics for Muzaffarnagar (IWMP, 2009) and reports of local Krishi Vigyan Kendra for the year 2014, sugarcane, wheat, rice are the principal crops for the studied region. In a gradual rising trend, sugarcane cultivation area has increased to>90% of the cultivated area, whereas the wheat and paddy are cultivated on around 57% and 10% of the remaining land, respectively. Also, the cropping intensity for the district is 153.2%. However, the results suggest that even for retrieving maximum profit from the land (i.e., under MIR scenario) which would essentially be at the cost of more water consumption, a maximum of about 20% of the land area can be allocated for sugarcane cultivation. The inclusion of fallow land in the cropping sequences is also an agricultural management practice followed by the farmers from centuries for restoring the soil's crop yield ability primarily by improving soil's nutrient, organic matter, and water concentrations (Nadeem et al., 2019).

4.4. Development of an interactive web-based dashboard for farmers

Interpreting the results by referring to the obtained raster's may be challenging due to the fixed resolution of the map. Further, to identify the user's location on the map, geo-referencing has to be performed. Since the outcomes of this study is a great interest to farmers, providing an accessible solution would help in the on-ground implementation of results. Considering this, for easy interpretation of the results, a web-based dashboard is developed and is made available to the stakeholders. The developed dashboard can be accessed at https://harishppuppala43.github.io/croprotation/.

Fig. 7 presents the snapshot of the developed dashboard. User location is a dynamic variable, while the data used for analysis is static. The dashboard tracks the geolocation of the user and overlays the location on the map, which helps in the easy interpretation of the type of practice that farmer can adopt. Flexibility is provided to the user to turn on and off the layers to navigate between different layers.

4.5. Future work and applicability of the proposed DST on a global level

The study performs advanced analysis on multi-layered spatial data using GIS and remote sensing technology. The analysis employed an array of flexible mathematical tools & models (GIS, SPC, and constrained optimization) to attain a sustainable crop-rotation pattern. The uncertainties associated with stakeholders and input data is adequately dealt by infusion of fuzzy logic in the DST. The DST and dashboard offers a flexible interface to users to modify/add/delete input datasets or cropping patterns. Thus, the DST is universally applicable provided the availability of fundamental datasets (Table 1) and knowledge on the local agricultural system. The primary focus of this study was to propose a DST that can aid in suggesting cost-effective and environmentally sustainable cropping pattern. Therefore, the crop yield or crop production was not specifically considered as one of the optimization objectives. However, the consideration of yield prediction is also of great importance as it ensures and supports policy making for national food security and thus can be deliberated by future studies. The present research could be enhanced by using higher resolution datasets, and more scientifically designed cropping sequences for a greater number of crops and criteria. Artificial intelligence and cutting-edge optimization techniques such as genetic algorithm-based optimization, ant colony optimization, particle swarm optimization, etc., could also be implemented for attaining better sustainable solutions.

5. Conclusions

Inappropriate decisions related to agricultural land-use planning can have severe repercussions on soil health, water resources, food security, and economic crisis, which are being perceived over time. The current research offers a DST that scientifically allocates the crop rotation patterns to potential land parcels by evaluating land's strengths and weaknesses as per the specific crop's requirements using an integrated application of geoinformatics, stochastic pairwise comparison, and constrained optimization. The approach enables management of a wide variety of multispatial data under the fields of climatology, topography, and soil properties, generates classified criterion layers, standardizes, and overlays them. The research could be further strengthened by incorporating micronutrients in addition to macronutrients under soil properties field. The use of SPC is especially advantageous in providing flexibility to the DMs in providing their opinions while ranking the criteria. Further, the infusion of constraint optimization assists farmers and other stakeholders in subjective selection of crop rotation patterns based on profit maximization and water conservation. However, consideration of other important factors like market demand of crops and man-hours per hectare required for different crops could enhance the cropping pattern selection decision. The analytical study of three scenarios revealed that the annual average cultivation increment and decrement of maize, mustard, and sugarcane crops describe the profit as well as water conservation potential of the agricultural land in Muzaffarnagar, respectively. The results also highlight many important facts such as: (1) Even for retrieving maximum profits, not more than 20% of land should be allocated for sugarcane cultivation (90% at present); (2) Mustard, potato, maize, and sorghum also have good cultivation potential which are currently not given due consideration in Muzaffarnagar. The inclusion of interactive web-based dashboard sets a common

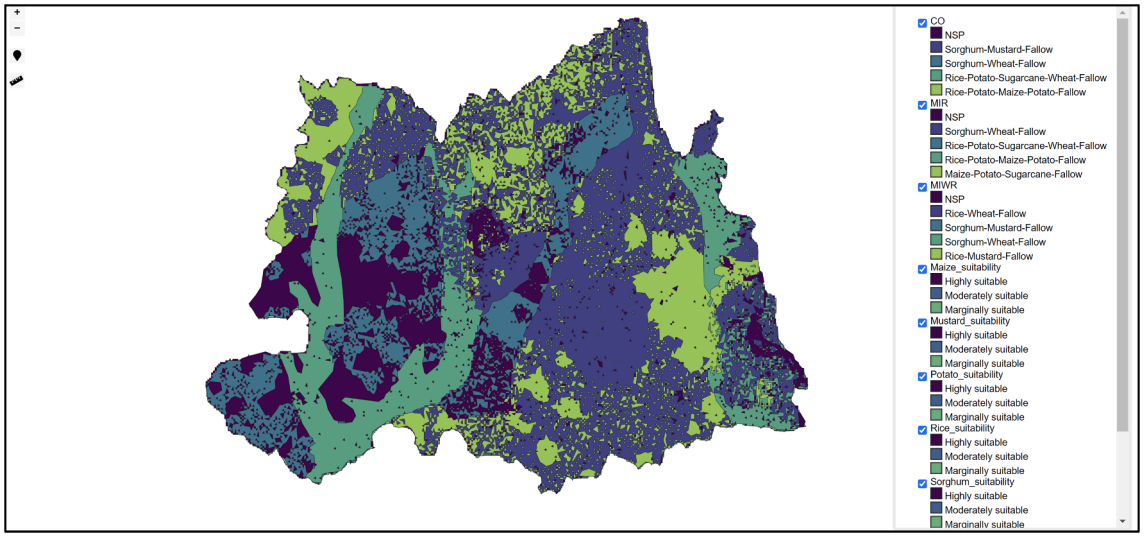


Fig. 7. Schematic illustration of the web-based dashboard.

interface for farmers, planners, and government officials where the knowledgebase, opinions, and conflicts related to the cropping pattern selection could be mutually shared. Thus, the proposed DST is a robust framework which simultaneously considers multitudes of factors: soil nutrients, terrain characteristics, crop requirements, hydrology, economics and promotes the sustainable agricultural land-use planning through field scale implementation of optimized crop rotation patterns.

CRediT authorship contribution statement

Shubham Aggarwal: Conceptualization, Data curation, Methodology, Software, Writing – original draft, Investigation, Visualization. **Rallapalli Srinivas:** Supervision, Conceptualization, Data curation, Methodology, Software, Writing – review & editing, Investigation, Visualization. **Harish Puppala:** Data curation, Visualization, Investigation, Writing – review & editing. **Joe Magner:** Writing – review & editing, Conceptualization, Visualization, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2022.107213.

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